SQL Exercise2: Aggregate Functions & Grouping.

1. COUNT ( ) Function:

```
SELECT  COUNT (ID) AS Number_of_employees
FROM    employee ;
```

← Output

Number_of_employees
10

2. SUM ( ) Function

```
SELECT  department,
        SUM (salary) AS Total_salary
FROM    employee
WHERE   department = 'IT' ;
```

OUTPUT

department	Total_salary
IT	220 000

3. AVG ( ) Function:

```
SELECT  department,
        AVG(salary) AS Average_salary
FROM    employee
WHERE   department = 'HR' ;
```

Output

Average_salary
49 500

4. MIN() and MAX() function.

```
SELECT  MIN(salary) AS Lowest_salary,
        Max(salary) AS Highest_salary
FROM    employee;
```

output

Lowest_salary	Highest_salary
48 000	62 000.

5. GROUP BY statement

```
SELECT  department,
        SUM(salary) AS Total_salary
FROM    employees
GROUP BY department;
```

Output

department.	Total_salary
IT	220000
HR	99 000
Finance	119 000
Marketing	105 000

6. GROUP BY and COUNT()

```
SELECT  city,
        COUNT(ID) AS Total_no_per_city
FROM    employees
GROUP BY city;
```

6.
OUTPUT

City	Total_no_per_City
New York	2
Chicago	3
Los Angeles	2
San francisco	2
Houston	1

7. GROUP BY and ORDER BY

```
SELECT department,
       Avg(salary) As Average_salary
FROM employees
~~ORDER BY~~
GROUP BY department
ORDER BY Average_salary DESC
```

OUTPUT

department	Average_salary
Finance	59 000
IT	55 000
Marketing	52 000
HR	49 000

8. HAVING clause-

```
SELECT department, SUM(salary) AS Total_salary
FROM employee
GROUP BY department
HAVING SUM(salary) > 100 000;
```

output

~~department~~	Total - salary
~~Finance~~	119 000

department	Total - salary
IT	220 000
Finance	119 000
Marketing	105 000

9. Combining GROUP BY, HAVING, and ORDER BY

```
SELECT    City,
          COUNT (ID) AS number_of_employees
FROM      employee
GROUP BY  City
HAVING    COUNT (ID) > 1
ORDER BY  number_of_employees DESCENDING
```

output number_of_employees

City	number_of_employees
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10. Combining Aggregate functions

```
SELECT  department
        Avg (salary) AS highest_average_salary
FROM    employee
GROUP BY department
ORDER BY Avg (salary) DESC
LIMIT 1.
```

OUTPUT

department	highest_average_salary
finance	59 000